Real Analysis, Normed spaces and linear operators
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5.1 Normed spaces

We are familiar with normed vector spaces from MAT1125. Its basically a vector space which we equip with a notion of distance. But in MAT1125, we were very comfortable with finite dimensional vector spaces, which as a consequence, has a basis. We are now interested in infinite dimensional vector spaces, which equivalently, does not have a basis. In the next chapter, we will see how we can extend the theory of differentiation and linearization to normed spaces.

Definisjon 1: 5.1.1

Let $\mathbb{K}$ be a body and let V be a non-empty set equipped with two opearations

- Addition: which to any two elements $u, v \in V$ assigns a value $u + v \in V$
- Scalar multiplication: which to any element $u \in V$ and $\alpha \in \mathbb{K}$ assigns an element $\alpha u \in V$.

We call V a vector space over $\mathbb{K}$ if

1. $u + v = v + u \ \forall u, v \in V$
2. $(u + v) + w = u + (v + w) \ \ \forall u, v, w \in V$
3. $\exists 0_V \in V$ such that $u + 0_V = u \ \forall u \in V$
4. For each $u \in V$, there is an element $-u \in V$ such that $u + (-u) = 0$
5. $\alpha (u + v) = \alpha u + \alpha v \ \forall u, v \in V$ and $\forall \alpha \in \mathbb{K}$
6. $(\alpha + \beta) u = \alpha u + \beta u \ \forall u \in V$ and $\forall \alpha, \beta \in \mathbb{K}$
7. $\alpha (\beta u) = (\alpha \beta) u \ \forall u \in V$ and $\forall \alpha, \beta \in \mathbb{K}$.
8. $1u = u \ \forall u \in V$

Definisjon 2: 5.1.2

If U is a vector space over $\mathbb{K}$, a norm on U is a function $\|\cdot\| : V \to \mathbb{R}$ such that

1. $\|u\| \geqslant 0$ and $\|u\| = 0 \iff u = 0$
2. $\|\alpha u\| = |\alpha| \, \|u\| \ \ \forall \alpha \in \mathbb{K}, u \in U$
3. $\|u + v\| \leqslant \|u\| + \|v\| \ \ \forall u, v \in U$

We have for the space $U = C(X, \mathbb{R})$ the set of all continuous real valued functions on X, then U is a vector space over $\mathbb{K} = \mathbb{R}$ and

$$\|f\| = \sup\{|f(x)| : x \in X\}$$

which is a norm on U.

Proposisjon 1: 5.1.3

Let $(V, \|\cdot\|)$ be a normed space. Then

$$d(u, v) = \|u - v\|$$

is a metric on V.

Bevis 1: 5.1.3

- Positivity: We see that since $d(u, v) = \|u - v\|$, then $d(u, v) \geqslant 0$ and $d(u, v) = 0 \iff u - v = 0 \Rightarrow u = v$ which is the requirement for a metric to be zero.
- Symmetry: SInce $\|u - v\| = \|(-1)(v - u)\| = |-1| \|v - u\| = \|v - u\|$, then $d(u, v) = d(v, u)$
- Triangle inequality. We have that $d(u, v) = \|u - v\| = \|(u - w) + (w - v)\| \leqslant \|u - w\| + \|w - v\| = d(u, w) + d(w, v) \ \forall u, v, w \in U$

Remark that the inverse triangle inequality $|d(u, v) - d(u, w)| \leqslant d(w, v)$ also has its own version in normed spaces.

$$|\|u\| - \|v\|| \leqslant \|u - v\|.$$

Remark the L^p

Proposisjon 2: 5.1.4

Assume $(U, \|\cdot\|)$ is a normed space.

- If (x_n) is a sequence from U converging to x, then $(\|x_n\|)$ converges to $\|x\|$
- If (x_n) and (y_n) converge to x and y respectively, then $(x_n + y_n)$ converge to $x + y$
- If (x_n) is a sequence from U converging to x and (α_n) is a sequence from $\mathbb{K}$ converging to α, then $(\alpha_n x_n)$ converges to αx.

Bevis 2: 5.1.4

- Let (x_n) be a sequence in U convergint to x. This means $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $\forall n \geqslant N$, then $\|x_n - x\| < \varepsilon$. Since
$$|\|x_n\| - \|x\|| \leqslant \|x_n - x\|,$$
then $|\|x_n\| - \|x\|| < \varepsilon\ \forall n \geqslant N$. This implies that we can choose N large enough so that for all other preceding elements in the sequence, the distance between x_n and x will be $< \varepsilon$ away from eachother. This implies that
$$\|x_n\| \to \|x\|$$
as $n \to \infty$.

- Let (x_n) converge to x and (y_n) converge to y. Then $\forall \varepsilon_1 > 0\ \exists N_1 \in \mathbb{N}$ such that $\forall n \geqslant N_1, \|x_n - x\| < \varepsilon_1$. And $\forall \varepsilon_2\ \exists N_2 \in \mathbb{N}$ such that $\forall n \geqslant N_2, \|y_n - y\| < \varepsilon_2$.

  We are supposed to show that $(x_n + y_n)$ converges to $x + y$. Let $\varepsilon > 0$, find $N \in \mathbb{N}$ such that $\forall n \geqslant N$, we then get that $\|(x_n + y_n) - (x + y)\| < \varepsilon$.

  Observe that
$$\|(x_n + y_n) - (x + y)\| = \|(x_n - x) + (y_n - y)\| \leqslant \|x_n - x\| + \|y_n - y\|$$
  Since for n greater than N_1, we get that $\|x_n - x\| < \varepsilon_1$ and for n greater than N_2, we get that $\|y_n - y\| < \varepsilon_2$, then we can pick $N = max(N_1, N_2)$ and define $\varepsilon := \varepsilon_1 + \varepsilon_2$ and get as a result that $\forall n \geqslant N$, then
$$\|(x_n + y_n) - (x + y)\| \leqslant \|x_n - x\| + \|y_n - y\| < \varepsilon_1 + \varepsilon_2 = \varepsilon.$$
  Hence, I have found an N such that for all n larger than n, then the distance between $(x_n + y_n)$ and $x + y$ is less than ε away. Hence,
$$(x_n + y_n) \to x + y$$
  as $n \to \infty$.

- Let (x_n) converge to x in U and (α_n) converge to α in $\mathbb{K}$. Then $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $\|x_n - x\| < \varepsilon\ \forall n \geqslant N$ and $\forall \sigma > 0\ \exists N_1 \in \mathbb{N}$ such that $|\alpha_n - \alpha| < \sigma\ \forall n \geqslant N_1$. Im supposed to show that $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $\|\alpha_n x_n - \alpha x\| < \varepsilon\ \forall n \geqslant N$

  Let $\varepsilon > 0$, and N great enough such that both $\|x_n - x\| < \varepsilon$ and $|\alpha_n - \alpha| < \delta$
$$\|\alpha_n x_n - \alpha x\| = \|(\alpha_n x_n + \alpha x_n) - (\alpha x_n + \alpha x)\| \leqslant \|\alpha_n x_n - \alpha x_n\| + \|\alpha x_n - \alpha x\| < \delta \|x_n\| + |\alpha| \varepsilon$$
  Since δ and ε can be arbitrarily small, then $\|\alpha_n x_n - \alpha x\| \to 0$ as $n \to \infty$

Definisjon 3: 5.1.5

Two norms $\|\cdot\|_1, \|\cdot\|_2$ are equivalent if there are positive constanct C_1, C_2 such that $\forall x \in U$
$$\|x\|_1 \leqslant C_1 \|x\|_2$$
and
$$\|x\|_2 \leqslant C_2 \|x\|_1.$$

Proposisjon 3: 5.1.6

Assume $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent. Then

1. If a sequence (x_n) converges to x w.r.t one of the norms, its also convergent w.r.t the other norm
2. If a set is open, closed, compact w.r.t one norm, its also open, closed, compact w.r.t the other norm.
3. If X is a metric space, and a map $f : U \to X$ is continuous w.r.t one norm, its also continuous w.r.t the other norm. Likewise with $g : X \to U$ is continuous.

Teorem 1: 5.1.7

All norms on $\mathbb{R}^n$ are equivalent.

Bevis 3: 5.1.7

Suffices to show that all norms are equivalent w.r.t euclidean norm $\|\cdot\|$. We have that

$$\|x\| = \sqrt{\sum_{j=}^{n} x_i^2} \leqslant \sqrt{\sum_{j=1}^{n} \|x\|_\infty^2} = n \|x\|_\infty .$$

And

$$\|x\|_\infty = \sup_k |x_k| \leqslant \sqrt{\sum_{k=1}^{n} x_k^2}$$

so the supremum norm and euclidean norm are equivalent.

Let $|\cdot|$ be another norm. We must show there are constants c_1 and c_2 such that

$$|x| \leqslant c_1 \|x\|$$

and

$$\|x\| \leqslant c_2 |x|$$

To prove the first inequality, let $(e_1, ..., e_n)$ be a basis for $\mathbb{R}^n$, and put

$$B = max \{|e_1|, ..., |e_n|\}$$

So for $x \in \mathbb{R}^n$ we have that

$$x = \sum_{j=1}^{n} x_j e_j.$$

Thus

$$|x| = \left|\sum_{j=1}^{n} x_j e_j\right| \leqslant \sum_{j=1}^{n} |x_k| |e_k| \leqslant \sum_{j=1}^{n} |x_j| B = B (|x_1| + ..., + |x_n|) \leqslant nBmax_j \{x_k\}$$

And since $max_j \{x_j\} = \sqrt{max_j |x_j|^2} \leqslant \sqrt{\sum_{j=1}^{n} x_j^2} = \|x\|$

Proposisjon 4: 5.1.8

If the spaces $(V_1, \|\cdot\|_1), ..., (V_m, \|\cdot\|_n)$ are complete, so is their product $(V, \|\cdot\|)$

Bevis 4: 5.1.8

We have that a normed space is complete if the metric space (U, d) where $d(x, y) := \|x - y\|$ is complete. We have that $V = V_1 \times ... \times V_n$ where $(V_1, \|\cdot\|_1), ..., (V_n, \|\cdot\|_n)$ are complete.

This means that if $(x_n)_i$ is a cauchy sequence in V_i, then $\forall \varepsilon_i > 0 \ \exists N_i \in \mathbb{N}$ such that $\forall k, j \geqslant N_i$, then $\left\|(x_j - x_k)_i\right\|_i < \varepsilon_i$.

Im supposed to show that $V = V_1 \times ... \times V_n$ is complete. Let $\varepsilon > 0$, and $(x_n) \in V$ be a cauchy seuqnence in V.

$$(x_n) = ((x_n)_1, ..., (x_n)_n)$$

Since $(x_n)_i$ are all cacuhy seuquences in a complete metric space, they all converge. So let $N = \sup_j \{N_j\}$ and $\varepsilon = \sup_j \{\varepsilon_j\}$. We then get that $\forall k, j \geqslant N$, then

$$\|x_n - x_m\| = \|x_j - x_k\|_1 + ... + \|x_j - x_k\|_n < \varepsilon_1 + ... + \varepsilon_n \leqslant \sum_{i=1}^{n} \varepsilon = n\varepsilon.$$

So we can just pick $\varepsilon' = \varepsilon/n$, and just use ε' as our new threshold, because this is still larger than 0. Hence, V is complete.

Oppgave 1: 5.1.5

Prove the inverese triangle inequality

$$|\|u\| - \|v\|| \leqslant \|u - v\| \ \ \forall u, v \in U.$$

Bevis 5: 5.1.5

Let $(U, \|\cdot\|)$ be a normed vector space. Then for any $u, v \in U$, then

$$\|u + v\| \leqslant \|u\| + \|v\|$$

Let $u, v \in U$. Then we can rewrite

$$\|u\| = \|(u - v) + v\| \leqslant \|u - v\| + \|v\|.$$

Then just subtract $\|v\|$ on both sides and get

$$\|u\| - \|v\| \leqslant \|u - v\| \Rightarrow |\|u\| - \|v\|| \leqslant \|u - v\|.$$

And the inverse triangle inequality is proved.

Oppgave 2: 5.1.10

Check that the product $(V, \|\cdot\|)$ of normed spaces $(V_1, \|\cdot\|_1), ..., (V_n, \|\cdot\|_n)$ really is normed space. Check that V is a linear space, and that $\|\cdot\|$ is a norm.

Bevis 6: 5.1.10

Let $V = V_1 \times V_2 \times ... \times V_n$. This is the set of all n-tuples. Let $x \in V$, then

$$x = (x_1, ..., x_n)$$

where $x_i \in V_i$ for $i = 1, ..., n$. To show that this is a linear space, prove that V satisfies all the axioms.

1. If $x, y \in V$, then $x = (x_1, ..., x_n)$ and $y = (y_1, ..., y_n)$. Then

$$x + y = (x_1 + y_1, ..., x_n + y_n) = (y_1 + x_1, ..., y_n + x_n) = y + x$$

2. If $\alpha \in \mathbb{K}$ and $x \in V$, then

$$\alpha x = (\alpha x_1, ..., \alpha x_n) = \alpha (x_1, ..., x_n)$$

   the rest are obvious.

And now to check that $\|\cdot\|$ is actually a norm. Let $x, y \in V$ and $\alpha \in \mathbb{K}$

1. $\|x\| = (\|x_1\|, ..., \|x_n\|)$, and this is equal to 0 if and only if $\|x_i\|_{i=1}^n = 0 \iff x_i = 0 \; \forall i = 1, ..., n$.
2. $\|\alpha x\| = (\|\alpha x_1\|, ..., \|\alpha x_n\|) = (|\alpha| \|x_1\|, ..., |\alpha| \|x_n\|) = |\alpha| (\|x_1\|, ..., \|x_n\|) = |\alpha| \|x\|$
3. $\|x + y\| = (\|x_1 + y_1\|, ..., \|x_n + y_n\|) \leqslant (\|x_1\| + \|y_1\|, ..., \|x_n\| + \|y_n\|) = (\|x_1\|, ..., \|x_n\|) + (\|y_1\|, ..., \|y_n\|) = \|x\| + \|y\|$

Hence $(V, \|\cdot\|)$ is a normed space.

Oppgave 3: 5.1.11

Prove proposition 5.1.8

5.1 ℓ^p-spaces

If we let (a_n) be a sequence in $\mathbb{R}$, then $a = (a_1, a_2, ...)$ which can be viewed as an infinite dimensional vector in. We instead let $a \in \ell^p$

5.1 Holders inequality

Teorem 2: Holders inequality

For $x, y \in \mathbb{F}^n$, we have that

$$\|xy\|_1 \leqslant \|x\|_p \|y\|_q$$

where $\frac{1}{p} + \frac{1}{q} = 1$ for $p, q \in [1, \infty]$ and

$$\|x\|_p = \left(\sum_{j=1}^n |x_j|^p \right)^{\frac{1}{p}}$$

We write

$$xy = (x_1 y_1, ..., x_n y_n)$$

Oppgave 4: 1

Show that $\|\cdot\|_1$ and $\|\cdot\|_\infty$ are norms on $\mathbb{R}^n$

Bevis 7: For $\|\cdot\|_1$

Let $u, v \in \mathbb{R}^n$ and $\alpha \in \mathbb{K}$. Then

1. $\|u\|_1 = \sum_{j=1}^n |u_j| \geqslant 0$ and is equal to 0 if $u = 0$
2. $\|\alpha u\|_1 = \sum_{j=1}^n |\alpha u_j| = \sum_{j=1}^n |\alpha| |u_j| = |\alpha| \sum_{j=1}^n |u_j| = |\alpha| \|u\|_1$
3. $\|u + v\|_1 = \sum_{j=1}^n |u_j + v_j| \leqslant \sum_{j=1}^n |u_j| + |v_j| = \sum_{j=1}^n |u_j| + \sum_{j=1}^n |v_j| = \|u\|_1 + \|v\|_1$

Bevis 8: For $\|\cdot\|_\infty$

Let $u, v \in \mathbb{R}^n$ and $\alpha \in \mathbb{K}$. Then

1. $\|u\|_\infty = max_{k=1,\dots,n} |u_k| \geqslant 0$ and is equal to 0 if the largest absolute value of each component of u is 0. This implies that $u = 0$
2. $\|\alpha u\|_\infty = max_{k=1,\dots,n} |\alpha u_k| = max_{k=1,\dots,n} |\alpha| |u_k| = |\alpha| max_{k=1,\dots,n} |u_k| = |\alpha| \|u\|_\infty$
3. $\|u+v\|_\infty = max_{k=1,\dots,n} |u_k + v_k| \leqslant max_{k=1,\dots,n} |u_k| + |v_k| \leqslant max_k |u_k| + max_k |v_k| = \|u\|_\infty + \|v\|_\infty$

For remaining exercises, assume $p \in (1, \infty)$

Oppgave 5: 2

Show that $\|\cdot\|_p$ satisfies the positivity and homogeneity conditions

Bevis 9: 2

1. We have for a $u \in \mathbb{R}^n$, then $\|u\|_p = \left(\sum_{j=1}^n |u_j|^p\right)^{\frac{1}{p}}$. If $u = 0$, then

$$\|u\|_p = \left(\sum_{j=1}^n |u_j|^p\right)^{\frac{1}{p}} = \left(\sum_{j=1}^n 0^p\right)^{\frac{1}{p}} = 0.$$

   Hence, $\|u\|_p \geqslant 0$ and $\|u\|_p = 0 \iff u = 0$

2. Next,

$$\|\alpha u\|_p = \left(\sum_{j=1}^n |\alpha u_j|^p\right)^{\frac{1}{p}} = \left(\sum_{j=1}^n |\alpha|^p |u_j|^p\right)^{\frac{1}{p}} = (|\alpha|^p)^{\frac{1}{p}} \left(\sum_{j=1}^n |u_k|^p\right)^{\frac{1}{p}} = |\alpha| \|u\|_p$$

Definisjon 4: 3

We say that $p, q \in (1, \infty)$ are conjugate exponents if

$$\frac{1}{p} + \frac{1}{q} = 1.$$

The conjugate exponent of 1 is ∞ .

Oppgave 6: 3

Let p, q be conjugate exponents.

1. Show that. if $p = q$, then $p, q = 2$
2. Show that if $p < 2$, then $q > 2$ and vice versa
3. Show that $q = \frac{p}{p-1}$

Bevis 10: 3 a)

If $p = q$, then

$$\frac{1}{q} + \frac{1}{p} = \frac{1}{q} + \frac{1}{q} = \frac{2}{q} = 1 \Rightarrow q = 2 = p$$

Bevis 11: 3 b)

Assume $p < 2$, then $\frac{1}{p} > \frac{1}{2}$. Let $k = \frac{1}{p}$. Then

$$\begin{aligned} k + \frac{1}{q} &= 1 \\ \frac{1}{q} &= 1 - k < \frac{1}{2}. \end{aligned}$$

So $\frac{1}{q} < \frac{1}{2}$ which implies that $q > 2$. Hence, if $p < 2$, then $q > 2$. And if $p > 2$, then $q < 2$.

Bevis 12: 3 c)

We have that

$$\begin{aligned} \frac{1}{p} + \frac{1}{q} &= 1 \\ p + q &= pq \\ p &= q\,(p-1) \\ q &= \frac{p}{p-1} \end{aligned}$$

Oppgave 7: 4

Prove Youngs inequality. If $p, q \in (1, \infty)$ are conjuage exponents, then

$$xy \leqslant \frac{|x|^p}{p} + \frac{|y|^q}{q} \ \forall x, y \in \mathbb{R}$$

- a) For a continuous function $f : \mathbb{R} \to \mathbb{R}$, the Legendre transform of f is the function $f^* : \mathbb{R} \to \mathbb{R}$ defined as

  $$f^*(y) := \sup_{z \in \mathbb{R}} (zy - f(z))$$

  Prove that $xy \leqslant f(x) + f^*(y) \ \forall x, y \in \mathbb{R}$
- b) Let $p, q \in (1, \infty)$ be conjugate exponents. Show that the Legendre transform of $f(x) := \frac{|x|^p}{p}$ is $f^*(y) = \frac{|y|^q}{q}$
  Hint: Think of $f^*(y)$ as the maximum of a function. How do you find maxima of functions on $\mathbb{R}$?
- Prove Young's inequality (1)

Bevis 13: 4 a)

Since $xy - f(x) \leqslant f^*(y)$ for all $x \in \mathbb{R}$, then

$$xy \leqslant f(x) + f^*(y) \ \forall x, y \in \mathbb{R}$$

Bevis 14: 4 b)

We start by observing that $p = \frac{q}{q-1}$ and $q = \frac{p}{p-1}$. Define the function $g(x) = xy - f(x)$. Start by optimizing $g(x)$, and get $\frac{d}{dx}\left(xy - \frac{|x|^p}{p}\right) = y - |x|^{p-1} = 0 \Rightarrow y = |x|^{p-1}$,

which implies that $x = y^{\frac{1}{p-1}} = y^{1-q}$, becuase $\frac{1}{p} + \frac{1}{q} = 1$.

$$q + p = pq \Rightarrow q\,(1+p) = -p \Rightarrow q = \frac{p}{p-1}$$
$$q - 1 = \frac{p}{p-1} - 1 \Rightarrow q - 1 = \frac{1}{p-1}$$

Also,

$$\frac{1}{q} = \frac{1}{\frac{p}{p-1}} = \frac{p-1}{p}$$

Substituting this back into our original equation, we get that

$$g(x) = y^{q-1}y - \frac{y^q}{p} = y^q\left(1 - \frac{1}{p}\right) = y^q\frac{1}{q}$$

Hence, $f^*(y) = \frac{|y|^q}{q}$

Bevis 15: 4 c)

Oppgave 8: 5

Problem 6.

Problem 6.

Problem 6. Next, we prove Hölder's inequality: If $u, v \in \mathbb{R}^n$, and $p, q \in [1, \infty]$ are conjugate exponents, then

$$\|uv\|_1 \leqslant \|u\|_p \|v\|_q. \tag{2}$$

(Here, uv is the vector $uv = (u_1v_1, \ldots, u_nv_n)$.)

(a) Prove the special case $p = 1$, $q = \infty$.

Bevis 16: 6 a)

If $p = 1$ and $q = \infty$, then

$$\|uv\|_1 = \sum_{k=1}^{n} |u_kv_k| = \sum_{k=1}^{n} |u_k|\,|v_k| \leqslant \sum_{k=1}^{n} |u_k|\,\|v\|_\infty = \|u\|_1\,\|v\|_\infty$$

hence, $\|uv\|_1 \leqslant \|u\|_1\,\|v\|_\infty$

(b) We now assume $p, q \in (1, \infty)$. Prove Hölder's inequality in the case where either $u = 0$ or $v = 0$.

Bevis 17: 6 b)

Let $u = 0$, then $u = (0, 0, ..., 0)$

$$\|uv\|_1 = \sum_{k=1}^{n} |u_kv_k| = \sum_{k=1}^{n} |u_k|\,|v_k| = \sum_{k=1}^{n} 0\,|v_k| = 0$$

(c) Assume now that $u, v \neq 0$. Explain each step in the following computation:

$$\frac{|uv|_1}{|u|_p\,|v|_q} = \frac{1}{|u|_p\,|v|_q}\sum_{i=1}^{n}|u_iv_i| = \sum_{i=1}^{n}\frac{|u_i|}{|u|_p}\frac{|v_i|}{|v|_q} \leqslant \sum_{i=1}^{n}\frac{|u_i|^p}{p\,|u|_p^p} + \frac{|v_i|^q}{q\,|v|_q^q}$$

$$= \frac{|u|_p^p}{p\,|u|_p^p} + \frac{|v|_q^q}{q\,|v|_q^q} = 1.$$

Bevis 18: 6 c)

The only thing we need to note here, is that in the third step, we set

$$x = |u_k| / \|u\|_p, y = |v_k| / \|v\|_q$$

and get that $xy = \frac{|u_k|}{\|u\|_p} \cdot \frac{|v_k|}{\|v\|_q} \leqslant \frac{1}{p}\left(\frac{|u_k|}{\|u\|_p}\right)^{\frac{1}{p}} \frac{1}{q}\left(\frac{|v_k|}{\|v\|_q}\right)^{\frac{1}{q}}$

At the end, we get that $\frac{1}{p} + \frac{1}{q} = 1$ because

$$\sum_{k=1}^{n} |u_k|^p = \left(\sum_{k=1}^{n} |u_k|^p\right)^{\frac{1}{p}\cdot p} = \|u\|_p^p$$

(d) Use the computation in the previous step to conclude Hölder's inequality.

5.2 Infinite sums and bases

A finite set of elements in V, is called a basis if all elements in V can be written as a *linear combination* of them. That is, given $\{u_1, ..., u_n\}$ a basis for V, then if $x \in V$

$$x = \alpha_1 u_1 + ... + \alpha_n u_n$$

for *unique* $\alpha_1, ..., \alpha_n$.

Since we are now dealing with vector spaces that are not finite, we run into a problem. We need to make sense of infinite sums in normed spaces.

Definisjon 5: 5.2.1

If $(u_k)_{k=1}^{\infty}$ is a sequence of elements in a normed space, we define the infinite sum $\sum_{k=1}^{\infty} u_k$ as the limit of the partial sums

$$S_n = \sum_{k=1}^{n} u_k$$

provided this element exists. i.e.

$$\sum_{k=1}^{\infty} u_k = \lim_{n\to\infty} \sum_{k=1}^{n} u_k.$$

When the limit exists, the series converges, otherwise, it diverges.

We can now extend the notion of a basis

Definisjon 6: 5.2.2

Let $(e_k)_{k=1}^{\infty}$ be a sequence of elements in a normed vector space V. We say that (e_k) is a basis for V if for each $x \in V$ there is a unique sequence (α_k) from $\mathbb{K}$ such that

$$x = \sum_{k=1}^{\infty} \alpha_k e_k$$

This doesnt mean all normed spaces have a basis. Some are uncountably infinite, which means not all elements can be reached from a countable set.

Oppgave 9: 5.2.1

Prove that the set (e_k) in example 1 really is a basis for c_0

Bevis 19: 5.2.1

We have that c_0 is the set of all sequences $x = (x_k)$ of real numbers such that $\lim_{k\to\infty} x_k = 0$. We have that $e_k = (0, 0, ..., 0, 1, 0, ..., 0, ...)$ be the sequence that is 1 at element number k and 0 elsewhere.

Let $(x_k) \in c_0$, then $\lim_{k\to\infty} x_k = 0$.

Oppgave 10: 5.2.2

Show that if a normed vector space V has a basis, then its also separable

Bevis 20: 5.2.2

A vector space is separable if it has a dense and countable subset. If a normed vector space has a basis (e_k), then for all $x \in V$, then there exists a unique sequence (α_k) such that

$$x = \sum_{k=}^{\infty} \alpha_k e_k$$

Let $x \in V$, and let (e_k) be a basis for V, with the unique sequence being (α_k). Then $x = \sum_{k=1}^{\infty} \alpha_k e_k$. Or

$$\left\| x - \sum_{k=1}^{\infty} \alpha_k e_k \right\| = \lim_{n\to\infty} \left\| x - \sum_{k=1}^{n} \alpha_k e_k \right\|$$

Let our subset be $D := \{y \in V : y = \sum_{k=1}^{n} q_k e_k, q_k \in \mathbb{Q}\ \forall k \in \mathbb{N}\}$. This is a countable subset of V, because its the sum of finite elements in a countable set. Note that $\mathbb{Q}$ is countable.

I need now to show that D is dense. This happens if for all $x \in V, \varepsilon > 0$, then there exists an element $y \in D$ such that

$$\|x - y\| < \varepsilon$$

As we have $x \in V$, then $x = \sum_{k=1}^{\infty} \alpha_k e_k$. So we can find an $N \in \mathbb{N}$ such that $\left\| x - \sum_{k=1}^{N} \alpha_k e_l \right\| < \frac{\varepsilon}{2}$. For $(q_n) \in \mathbb{Q}$, we can find $|\alpha_k - q_k| < \frac{\varepsilon}{2N\|e_k\|}$. We then get that

$$\left\| \sum_{k=1}^{N} \alpha_k e_k - \sum_{k=1}^{N} q_k e_k \right\| \leqslant \sum_{k=1}^{N} \|(\alpha_k - q_k)\, e_k\| = \sum_{k=1}^{N} |\alpha_k - q_k|\, \|e_k\| < \sum_{k=1}^{N} \|e_k\| \frac{\varepsilon}{2N\,\|e_k\|} = \frac{\varepsilon}{2} \sum_{k=1}^{N} \frac{1}{N} = \frac{\varepsilon}{2}$$

Now, since we are supposed to prove that $\forall x \in V, \varepsilon > 0\ \exists y \in D$ such that $\|x - y\| < \varepsilon$, we see that

$$\|x - y\| = \left\| x - \sum_{k=1}^{N} \alpha_k e_k + \sum_{k=1}^{N} \alpha_k e_k \right\| = \left\| \left(x - \sum_{k=1}^{N} \alpha_k e_k \right) + \left(\sum_{k=1}^{N} \alpha_k e_k \right) \right\| \leqslant$$

$$\left\| x - \sum_{k=1}^{N} \alpha_k e_k \right\| + \left\| \sum_{k=1}^{N} \alpha_k e_k - y \right\| < \varepsilon/2 + \varepsilon/2 = \varepsilon$$

Hence, D is both countable, dense, and a subset of V, which implies that V is separable, when it has a basis.

Does the same apply for when V is finite dimensional? Since V itself is finite, its definetly countable, and one can also prove it has a dense subset I imagine.

Oppgave 11: 5.2.3

ℓ_1 is the set of all sequences $x = (x_k)$ of real numbers such that $\sum_{k=1}^{\infty} |x_k|$ converges.

a) show that $\|x\| = \sum_{k=1}^{\infty} |x_k|$ is a norm on ℓ_1

Bevis 21: 5.2.3 a)

i) obvious

ii) obvious

iii) Let $x, y \in \ell_1$, then $\|x\| = \sum_{k=1}^{\infty} |x_k|$ and $\|y\| = \sum_{k=1}^{\infty} |y_k|$. THen

$$\|x+y\| = \sum_{k=1}^{\infty} |x_k + y_k| = \lim_{n\to\infty} \sum_{k=1}^{n} |x_k + y_k| \leqslant \lim_{n\to\infty} \sum_{k=1}^{n} |x_k| + |y_k| = \lim_{n\to\infty} \left(\sum_{k=1}^{n} |x_k| + \sum_{k=1}^{n} |y_k| \right)$$

$$= \lim_{n\to\infty} \sum_{k=1}^{n} |x_k| + \lim_{n\to\infty} \sum_{k=1}^{n} |y_k| = \|x\| + \|y\|$$

b) Show that the set (e_k) as used in exc 5.2.1 is a basis for ℓ_1

Bevis 22: 5.2.3 b)

We have that a sequence (e_k) is a basis if for all $x \in \ell_1$ there exists a unique sequence (α_k) such that $x = \sum_{k=1}^{\infty} \alpha_k e_k$.

Let $\alpha_k = x_k$. Then

$$x - \sum_{k=1}^{N} x_k e_k = (0, ..., 0, x_{N+1}, ...).$$

This means that

$$\left\| x - \sum_{k=1}^{N} x_k e_k \right\| = \sum_{k=N+1}^{\infty} |x_k|.$$

Since $x \in \ell_1$, then $\sum_{k=1}^{\infty} |x_k|$ converges. This means that for large enough N, then

$$\sum_{k=1}^{\infty} |x_k| - \sum_{k=1}^{N} |x_k| < \varepsilon.$$

Which implies that $\sum_{k=N+1}^{\infty} |x_k| < \varepsilon$. So (x_k) is that unique sequence that we are looking for, with our basis (e_k) from exercise 5.2.1.

Now I have to prove that this sequence (x_k) is unique. To do this, I let (a_k) be another sequence, and find that $\left\|\sum_{k=1}^{\infty} x_k e_k - \sum_{k=1}^{\infty} a_k e_k\right\| = \lim_{n\to\infty} \left\|\sum_{k=1}^{n} x_k e_k - \sum_{k=1}^{n} a_k e_k\right\|$ does not converge.

Let (a_k) be another sequence. Then for at least one i, then $|x_i - a_i| > \varepsilon$. So $\|x - \sum_{k=1}^{n} a_k e_k =\| = \|(x - 1 - a - 1, ..., x_n - a_n, x_{n+1}, ...)\| = \sum_{k=1}^{n} |x_k - a_k| + \sum_{k=n+1}^{\infty} |x_k|$. Since $\sum_{k=1}^{n} |x_k - a_k| \geqslant \varepsilon$ and $\sum_{k=n+1}^{\infty} |x_k| < \varepsilon$, then $\|x - \sum_{k=1}^{n} a_k e_k\| > \varepsilon$. So $x \neq \sum_{k=1}^{n} a_k e_k$, which implies that (e_k) is not a basis for ℓ_1. And (x_k) is a unique sequence such that

$$x = \sum_{k=1}^{\infty} x_k e_k.$$

c) Show that ℓ_1 is complete

Bevis 23: 5.2.3 c)

Let $x \in \ell_1$, then $\|x\| = \sum_{k=1}^{\infty} |x_k| < \infty$. We have that $x = (x_n)$ is actually a seuquence. A sequence in ℓ_1 is (x_n^r), which can be viewed as an array

$$\begin{aligned} x_n^1 &= \left(x_1^1, x_2^1, \dots\right) \\ x_n^2 &= \left(x_1^2, x_2^2, \dots\right) \\ &\vdots \end{aligned}$$

Let (x_n^r) be a cauchy sequence, then $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $\forall i, j \geqslant N$, then $\left\|x_n^i - x_n^j\right\| = \sum_{k=1}^{\infty} \left|x_k^i - x_k^j\right| < \varepsilon$.

We have to prove that (x_n^r) converges to $x \in \ell_1$. Fix $N \in \mathbb{N}$. We now look at the sequences, and see that

$$\left|x_n^i - x_n^j\right| \leqslant \sum_{k=1}^{\infty} \left|x_k^i - x_k^j\right| < \varepsilon.$$

So for every fixed value of n, then (x_n^r) is a cauchy sequence i.e. it converges to something. Say (x_n^r) converges to S_n. So

$$\lim_{r \to \infty} x_n^r = S_n$$

We want to show that $\sum_{k=1}^{\infty} |x_k^r - S_n| < \varepsilon$.

Since $\sum_{k=1}^{\infty} \left|x_k^i - x_k^j\right| < \varepsilon$, then $\sum_{k=1}^{M} \left|x_k^i - x_k^j\right| < \frac{\varepsilon}{2}$. Let $j \to \infty$ and get

$$\sum_{k=1}^{M} \left|x_k^i - S_n\right| \leqslant \frac{\varepsilon}{2}.$$

Now let $M \to \infty$ and get

$$\sum_{k=1}^{\infty} \left|x_k^i - S_n\right| \leqslant \frac{\varepsilon}{2} < \varepsilon.$$

Now we want to prove that (S_n) is in ℓ_1. We have that (S_n) is cauchy, which implies that its bounded, since an element in ℓ_1 can only take real values. So $\exists R \in \mathbb{R}$ such that $\|x_n^r\| \leqslant R$, which implies that $\sum_{k=1}^{M} |x_k^r| \leqslant R$. Let $r \to \infty$ and get $\sum_{k=1}^{M} |S_k| \leqslant R$. Let $M \to \infty$ and get

$$\sum_{k=1}^{\infty} |S_k| \leqslant R.$$

Hence, $(S_n) \in \ell_1$ and ℓ_1 is complete

Oppgave 12: Recap from MAT1125

Use the theory from MAT1125 to show that the functions $e_k = e^{ikt}, k \in \mathbb{Z}$ constitute a basis for the space $C(\mathbb{T}, \mathbb{C})$ of complex-valued, continuous function on the torus, equipped with the L2 norm

$$\|f\| = \frac{1}{2\pi} \int_0^{2\pi} |f(t)|^2 \, dt$$

Bevis 24

We have that the space of continuous functions from the torus to complex values, $C(\mathbb{T}, \mathbb{C})$ is an infinite dimensional space. This space, as we have shown in this chapter, is that this can have a basis. A basis means an orthonormal basis in the Hilbert space sense: a set is both orthonormal and complete, meaning every f can be written as

$$f = \sum_{k \in \mathbb{Z}} \hat{f}_k e_k$$

with convergence in the L2 norm. We have that a sequence (e_k) is a basis for a normed vector space if for all $x \in V$ there exists a unique sequence (α_n) such that

$$x = \sum_{k=1}^{\inf} \alpha_k e_k$$

Or $x = \lim_{n \to \infty} \sum_{k=1}^{n} \alpha_k e_k$, which means that given $\varepsilon > 0$, then $\forall n \geqslant N$,

$$\left\| x - \sum_{k=1}^{n} \alpha_k e_k \right\| < \varepsilon$$

So when we are given our L2 norm, we need to prove convergence in this norm. From MAT1125, we proved that for every $f \in C(\mathbb{T}, \mathbb{C})$, then the fourier series of f is given by

$$f_k = \sum_{k=-n}^{n} \hat{f}_k e_k$$

and this converges uniformly to f, see lemma 4.3.3.

According to theorem 6.4.3, then uniform convergence implies convergence in the L2 norm as well. This means that

$$\lim_{k \to \infty} \left\| f - \sum_{k=-n}^{n} \hat{f}_k e_k \right\|_{L^2} = 0$$

Hence, there exists a unique sequence $\left(\hat{f}_k\right)$ in $C(\mathbb{T}.\mathbb{C})$ such that for all x in $C(\mathbb{T}.\mathbb{C})$ then $x = \sum_{k \in \mathbb{Z}} \hat{f}_k e_k$.

5.3 Inner product spaces

Definisjon 7: 5.3.1

An inner product $\langle \cdot, \cdot \rangle$ on V over $\mathbb{K}$ is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{K}$ such that $\forall u, v, w, c \in V, \alpha \in \mathbb{K}$

i) $\langle u, v \rangle = \overline{\langle v, u \rangle}$

ii) $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$

iii) $\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$

iv) $\langle u, u \rangle \geqslant 0$ and $\langle u, u \rangle = 0 \iff u = 0$

Some other useful identities are

i) $\langle u, \alpha v \rangle = \overline{\alpha} \langle u, v \rangle$

ii) $\langle \alpha u, \alpha v \rangle = |\alpha|^2 \langle u, v \rangle$

iii) $\langle u + v, w + c \rangle = \langle u, w \rangle + \langle u, c \rangle + \langle v, w \rangle + \langle v, c \rangle$

Proposisjon 5: 5.3.2 Pyhtagoran theorem

For all $u_1, ..., u_n$ orthogonal in V,

$$\|u_1 + ... + u_n\|^2 = \|u_1\|^2 + ... + \|u_n\|^2$$

Bevis 25: 5.3.2

We have that $\|u_1 + ... + u_n\|^2 = \langle u_1 + ... + u_n, u_1 + ... + u_n \rangle = \sum_{k,j=1}^{n} \langle u_k, u_j \rangle = \sum_{k=1}^{n} \langle u_k, u_k \rangle = \sum_{k=1}^{n} \|u_k\|^2 = \|u_1\|^2 + ... + \|u_n\|^2$ where we have used orthogonality.

Proposisjon 6: 5.3.3

Assume u and v are nonzero elements of V. THen the *projection* p of u on v is given by

$$p = \frac{\langle u, v \rangle}{\|v\|^2} v$$

Bevis 26: 5.3.3

Since p is parallell to v, it must be on the form $p = \alpha v$. To determine α, we note that in order for $u - p$ to be orthogonal to v, we must have $\langle u - p, v \rangle = 0$

$$\langle u - p, v \rangle = \langle u, v \rangle - \langle p, v \rangle = \langle u, v \rangle - \langle \alpha v, v \rangle = \langle u, v \rangle - \alpha \|v\|^2$$

which implies that $\alpha == \frac{\langle u,v \rangle}{\|v\|^2}$ and hence,

$$p = \alpha v = \frac{\langle u, v \rangle}{\|v\|^2} v$$

Proposisjon 7: 5.3.4 Cauchy Schwarz inequality

for all $u, v \in V$

$$|\langle u, v \rangle| \leqslant \|u\| \|v\|$$

Bevis 27: 5.3.4

$$\|u\|^2 = \|u - p + p\|^2 = \|u - p\|^2 + \|p\|^2 \geqslant \|p\|^2 .$$

We have from 5.3.3 that $\|p\|^2 = \langle \alpha v, \alpha v \rangle = |\alpha|^2 \langle v, v \rangle = \frac{|\langle u,v \rangle|^2}{\|v\|^4} \|v\|^2$. So $\|p\| = \frac{|\langle u,v \rangle|}{\|v\|^2}$. Hence, we get that

$$\|u\|^2 \geqslant \frac{|\langle u, v \rangle|^2}{\|v\|^2} \Rightarrow |\langle u, v \rangle| \leqslant \|u\| \|v\| .$$

We can take roots, because the norms are nonzero.

Proposisjon 8: 5.3.5 Triangle inequality

for all $u, v \in V$

$$\|u + v\| \leqslant \|u\| + \|v\|$$

Bevis 28: 5.3.5

$\|u + v\|^2 = \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle = \langle u, u \rangle + \langle u, v \rangle + \overline{\langle u, v \rangle} + \langle v, v \rangle = \|u\|^2 + 2Re\left(\langle u, v \rangle\right) + \|v\|^2 \leqslant \|u\|^2 + 2\|u\| \|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$ where we have used cauchy inequality on

$$Re\left(\langle u, v \rangle\right) \leqslant |\langle u, v \rangle| \leqslant \|u\| \|v\|$$

Proposisjon 9: 5.3.6

If $\langle \cdot, \cdot \rangle$ is an inner product on V, then

$$\|u\| = \sqrt{\langle u, u \rangle}$$

defines a norm in V.

Proposisjon 10: 5.3.7

Let V be an inner product space

i) If (u_n) is a sequence in V converging to u,, then the sequence $(\|u_n\|)$ converges to $\|u\|$

ii) If the series $\sum_{k=0}^{\infty} w_k$ converges in V, then

$$\left\| \sum_{k=0}^{\infty} w_k \right\| = \lim_{n\to\infty} \left\| \sum_{k=0}^{n} w_k \right\|$$

iii) If (u_n) converges to u, then the sequence $\langle u_n, v\rangle$ converges to $\langle u, v\rangle$ for all $v \in V$

$$\lim_{n\to\infty} \langle u_n, v\rangle = \langle u, v\rangle$$

iv) If $\sum_{k=0}^{\infty} w_k$ converges in V, then

$$\left(\sum_{k=0}^{\infty} \langle w_k, v\rangle \right) = \sum_{k=0}^{\infty} \langle w_k, v\rangle$$

p

Oppgave 13: 5.3.8

Assume $(u_n) \to u$ and $(v_n) \to v$. Show that $\langle u_n, v_n\rangle \to \langle u, v\rangle$

Bevis 29: 5.3.8

We use the cauchy-schwartz inequality

$$|\langle x, y\rangle| \leqslant |x|\,|y|$$

given $\varepsilon > 0$, we have to find $N \in \mathbb{N}$ such that for all preceding elements in the sequence $(|\langle u_n, v_n\rangle|)$ then $|\langle u_n, v_n\rangle - \langle u, v\rangle| < \varepsilon$.

Given $\varepsilon_1 > 0, \varepsilon_2 > 0$ we have that for all $n \geqslant N_1 \in \mathbb{N}$ then $\|u - u_n\| < \varepsilon_1$ and for all $n \geqslant N_2 \in \mathbb{N}$ then $\|v - v_n\| < \varepsilon_2$.

So $|\langle u_n, v_n\rangle - \langle u, v\rangle| = |\langle u_n, v_n\rangle - \langle u, v_n\rangle + \langle u, v_n\rangle - \langle u, v\rangle| \leqslant |\langle u_n, v_n\rangle - \langle u, v_n\rangle| + |\langle u, v_n\rangle - \langle u, v\rangle| = |\langle u_n - u, v_n\rangle| + |\langle v_n - v, u\rangle| \leqslant \|u_n - u\|\,\|v_n\| + \|v_n - v\|\,\|u\| < \varepsilon_1 \|v_n\| + \varepsilon_2 \|u\|$. Since u is a fixed vector, we can bound it, and since v_n is a convergent sequence of vectors, we can also bound it for large enough n. Hence,

$$|\langle u_n, v_n\rangle - \langle u, v\rangle| < \varepsilon := \varepsilon_1 C + \varepsilon_2 M$$

Oppgave 14: 5.3.9

Show that if the norm $\|\cdot\|$ defined from an inner product $\|u\| = \langle u, u\rangle^{\frac{1}{2}}$, we have the parallellogram law

$$\|u + v\|^2 + \|u - v\|^2 = 2\,\|u\|^2 + 2\,\|2\|^2$$

for all $u, v \in V$. Show that the norms on $\mathbb{R}^2$ defined by $\|(x, y)\| = max\,\{|x|, |y|\}$ and $\|(x, y)\| = |x| + |y|$ do not come from inner products

Bevis 30: 5.3.9

We have that $\|u+v\|^2 = \langle u+v, u+v\rangle$ and $\|u-v\| = \langle u-v, u-v\rangle$.Then

$$\langle u+v, u+v\rangle = \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle = \|u\|^2 + 2\langle u,v\rangle + \|v\|^2$$

and

$$\langle u-v, u-v\rangle = \langle u,u\rangle - \langle u,v\rangle - \langle v,u\rangle + \langle v,v\rangle = \|u\|^2 - 2\langle u,v\rangle + \|v\|^2$$

then

$$\|u+v\|^2 + \|u-v\|^2 = \|u\|^2 + 2\langle u,v\rangle + \|v\|^2 + \|u\|^2 - 2\langle u,v\rangle + \|v\|^2 = 2\|u\|^2 + 2\|v\|^2.$$

Hence the parallellogram law.

For the inner product induced norms, assume that $u \in \mathbb{R}^2$ given by $u = (x,y)$, and that $|x| > |y|$. Then $\|u\| = max\{|x|, |y|\} = |x| \neq (|x| + |y|)^{\frac{1}{2}}$, and same argument for $\|u\| = |x| + |y|$

Oppgave 15: 5.3.10

Let $\{e_1, ..., e_n\}$ be an orthonormal set in an inner product space V. Show that the projection $P = P_{e_1,...,e_n}$ is linear in the sense that $P(\alpha u) = \alpha P(u)$ and $P(u+v)?P(u) + P(v)$

Bevis 31: 5.3.10

We have that for a vector space $u \in V$, then the projection onto the set of orthonormal vector $\{e_1, ..., e_n\}$ is given by

$$P_{e_1,...,e_n}(u) = \sum_{k=1}^{n} \langle u, e_k\rangle e_k.$$

So

$$P(\alpha u) = \sum_{k=1}^{n} \langle \alpha u, e_k\rangle e_k = \sum_{k=1}^{n} \alpha\langle u, e_k\rangle e_k = \alpha \sum_{k=1}^{n} \langle u, e_k\rangle e_k = \alpha P(u).$$

Second,

$$P(u+v) = \sum_{k=1}^{n} \langle u+v, e_k\rangle e_k = \sum_{k=1}^{n} (\langle u, e_k\rangle + \langle v, e_k\rangle) e_k = \sum_{k=1}^{n} \langle u, e_k\rangle e_k + \langle v, e_k\rangle e_k = P(u) + P(v).$$

Hence, the orthogonal projection is a linear operator.

Oppgave 16: 5.3.12

If S is a nonempty subset of an inner product space V, let

$$S^{\perp} = \{u \in V : \langle u, s\rangle = 0 \ \forall s \in S\}$$

a) Show that $S^{\perp}$ is a closed subspace of V

Bevis 32: 5.3.12 a)

We can prove a set is closed, if its complement is open We have that the set $S^{\perp} = \{u \in V : \langle u, s\rangle = 0 \ \forall s \in S\}$. So the complement would be

$$\left(S^{\perp}\right)^{\perp} = \{u \in V : \exists s \in S, \langle u, s\rangle \neq 0\}.$$

This means that we can at least one vector in S such that for all vectors $u \in V$ then $\langle u, s\rangle \neq 0$. In fact, this only vector is the zero vector. So if we choose a point in the neighbourhood of 0, we can find a radius, such that this neighborhood is completely contained within S.

b) Show that if $S \subseteq T$ then $T^{\perp} \subseteq S^{\perp}$.

Bevis 33: 5.3.12 b)

Assume that $t \in T^{\perp}$, then $\langle t, s \rangle = 0 \ \forall s \in T$. Since $S \subseteq T$, then also $\langle t, s^* \rangle = 0 \ \forall s^* \in S$. Hence, $\langle t, s^* \rangle = 0 \ \forall s^* \in S \Rightarrow t \in S^{\perp}$. Hence

$$T^{\perp} \subseteq S^{\perp}$$

5.4 Linear Operators

In linear algebra, we dealt with functions that are linear. This can be integration, differentiation, or matrices.

Definisjon 8: 5.4.1

Assume V and W are two vector spaces over $\mathbb{K}$. A function $A : V \to W$ is called linear if $\forall u, v \in V, \alpha \in \mathbb{K}$

i) $A(\alpha u) = \alpha A(u)$

ii) $A(u + v) = A(u) + A(v)$

ALSO, note that $A(\mathbf{0}) = A(0\mathbf{0}) = 0A(\mathbf{0}) = \mathbf{0}$ for all linear operators.

Definisjon 9: 5.4.2

Assume $(V, \|\cdot\|)$ and $(W, \|\cdot\|)$ are two normed vector spaces. A linear operator $A : V \to W$ is bounded, if there is a constant $M \in R$ such that

$$\|Au\|_W \leqslant M \|u\|_V$$

for all $u \in V$

You should note that bounded linear operators, does not mean the same as we defined in metric spaces, where boundedness, means that its finitely stretched. For linear operators, boundedness means continuous. We shall see this later on.

Note that if $A(u) \neq 0$, then we can make $\|A(\alpha u)\|_W = |\alpha| \|A(u)\|_W$ which we can make as large as we possibly want. What this means, is that if we can only stretch a vector up to a limit, then the space itself is not a vector space.

The smallest constant M in the definition 5.4.2 is denoted $\|A\|$

$$\|A\| = \sup \left\{ \frac{\|Au\|_W}{\|u\|_V} : u \neq 0 \right\}$$

alternatively

$$\|A\| = \sup \{ \|Au\|_W : \|u\|_V = 1 \}$$

$\|A\|$ is called the *operator norm*. Note that $\|Au\|_W \leqslant \|A\| \|u\|_V$

We have seen that integration is linear. Define $A : V \to \mathbb{R}$ where $V = C([a, b], \mathbb{R})$ as

$$A(u) = \int_a^b u(x)dx.$$

A is a linear functional, while $B : V \to V$ defined as

$$B(u)(x) = \int_a^x u(t)dt$$

is a linear operator. Its easy to check this is linear. Difference between a linear functional and a linear operator, is that a functional is a function of x, it takes an input, and spits out an output. An opearator, is something we do to for example a function, or a vector, or a number. Taking the supremum of a set, is an operator.

Now look at differentiation: $D : V \to U$ where $U = C([0, 1], \mathbb{R})$ and $V = \{g : [0, 1] \to \mathbb{R} : \text{ g is differentiable with } g' \in U\}$ as

$$D(f) = f'$$

This is also a linear operator.

The A and B operators are also bounded, if we consider the usual supremum norm on V. To see this for B, note that

$$|B(u)(x)| = \left| \int_a^x u(t)dt \right| \leqslant \int_a^x |u(t)| \, dt \leqslant \int_a^x \|u\| \, dt = \|u\| (x - a) \leqslant \|u\| (b - a)$$

which implies that $\|B(u)\| \leqslant \|u\| (b - a)$ for all $u \in V$

Lemma 1: 5.4.3

A bounded linear operator A is uniformly continuous.

Bevis 34: 5.4.3

If $\|A\| = 0$, A is constant zero, and nothing to prove. For $\|A\| \neq 0$, we may given $\varepsilon > 0$ choose $\delta := \frac{\varepsilon}{\|A\|}$ and get that when $\|x - y\| < \delta$, then

$$\|A(x) - A(y)\| = \|A(x-y)\| \leqslant \|A\| \|x - y\| < \|A\| \frac{\varepsilon}{\|A\|} = \varepsilon$$

which shows that A is uniformly continuous.

Lemma 2: 5.4.4

If a linear operator A is continuous at $\mathbf{0}$, its bounded.

Bevis 35: 5.4.4

Assume that A is continuous at 0. Let $u = 0, \varepsilon = 1$ as in the definition of continuity. Then we find that there is a $\delta > 0$ such that

$$\|0 - v\| < \delta \Rightarrow \|A(0) - A(v)\| < 1$$

i.e.

$$\|v\| < \delta \Rightarrow \|Av\| < 1$$

Let $u \in V$ be arbitrary. Then if $v := \frac{\delta}{2\|u\|} u$ has norm

$$\|v\| = \left\| \frac{\delta}{2\|u\|} u \right\| = \frac{\delta}{2} < \delta.$$

This implies that $\|Av\| < 1$. But then

$$\|Au\| = \left\| A\left(\frac{2\|u\|}{\delta} v\right) \right\| = \frac{2\|u\|}{\delta} \|Av\| = c\|u\| \|Av\| < c\|u\| \cdot 1$$

So there is a $c \in \mathbb{R}$ such that

$$\|Au\| \leqslant c\|u\| \quad \forall u \in V$$

Teorem 3: 5.4.5

For linear operators $A : V \to W$ between normed spaces, the following are equivalent

i) A is bounded

ii) A is uniformly continuous

iii) A is continuous at $\mathbf{0}$

Definisjon 10: 5.4.6

Ig V and W are normed spaces, we let $\mathcal{L}(V, W)$ denote the set of all bounded linear maps $A : V \to W$.

Proposisjon 11: 5.4.7

If V and W are normed spaces, the operator norm is a norm on $\mathcal{L}(V, W)$

Bevis 36: 5.4.7

We must check the def of a norm.

i) $\|A\| \geqslant 0$ We must show that $\|A\|$ is greater than 0, and is equal to 0 $\iff$ A is zero.

$$\|A\| = \sup\left\{\frac{\|Au\|_W}{\|u\|_V} : u \neq 0\right\}$$

which is clearly non negative, because a norm of something nonnegative

ii) $\|\alpha A\| = |\alpha|\,\|A\|$ We have that

$$\begin{aligned}\|\alpha A\| = \sup\left\{\frac{\|\alpha Au\|_W}{\|u\|_V} : u \neq 0\right\} &= \sup\left\{\frac{|\alpha|\,\|Au\|_W}{\|u\|_V} : u \neq 0\right\} \\ &= |\alpha| \sup\left\{\frac{\|Au\|_W}{\|u\|_V} : u \neq 0\right\} = |\alpha|\,\|A\|\end{aligned}$$

iii) For $A, B \in \mathcal{L}(V, W)$ then $\|A + B\| \leqslant \|A\| + \|B\|$

$$\begin{aligned}\|A + B\| = \sup\left\{\frac{\|(A+B)(u)\|_W}{\|u\|_V} : u \neq 0\right\} &= \sup\left\{\frac{\|A(u) + B(u)\|_W}{\|u\|_V} : u \neq 0\right\} \\ \leqslant \sup\left\{\frac{\|A(u)\|_W}{\|u\|_V} + \frac{\|B(u)\|_W}{\|u\|_V} : u \neq 0\right\} &\leqslant \sup\left\{\frac{\|A(u)\|_W}{\|u\|_V} : u \neq 0\right\} + \sup\left\{\frac{\|B(u)\|_W}{\|u\|_V} : u \neq 0\right\} \\ &= \|A\| + \|B\|\end{aligned}$$

I will explain the last inequality conceptually. Say we have a set. We take the max of that set, its a single number. Lets say we split the set now, and take the max of each new separate sets. Then at least one of the sets must contain the previous max. But we also have to add the new max thats lower, but still a number. This is why

$$\sup(A + B) \leqslant \sup(A) + \sup(B)$$

Hence, the operator norm, is a norm on the set $\mathcal{L}(V, W)$

Teorem 4: 5.4.8

Assume V and W are two normed spaces. If W is complete, so is $\mathcal{L}(V, W)$

Bevis 37: 5.4.8

We must prove that any cauchy sequence $(A_n) \in \mathcal{L}(V, W)$ converges to an element $A \in \mathcal{L}(V, W)$- First observe for any $u \in V$, then

$$\|A_n(u) - A_m(u)\| = \|(A_n - A_m)(u)\| \leqslant \|A_n - A_m\| \|u\|$$

which implies that $(A_n(u))$ is a cauchy sequence in W. Since W is complete, the sequence converges to a point we shall call $A(u)$

$$A(u) = \lim_{n\to\infty} A_n(u) \ \forall u \in V$$

This defines a function A from V to W. We need to prove its a bounded linear operator, and that (A_n) converges to A in operator norm.

$$A(\alpha u) = \lim_{n\to\infty} A_n(\alpha u) = \lim_{n\to\infty} \alpha A_n(u) = \alpha \lim_{n\to\infty} A_n(u) = \alpha A(u)$$

and

$$A(u+v) = \lim_{n\to\infty} A_n(u+v) = \lim_{n\to\infty} A_n(u) + A_n(v) = \lim_{n\to\infty} A_n(u) + \lim_{n\to\infty} A_n(v) = A(u) + A(v)$$

where we have used that the A_n's are linear operators.

Next step is to prove A is bounded. Note that by the inverse triangle inequality of norms, $|\|A_n\| - \|A_m\|| \leqslant \|A_n - A_m\|$, which shows that $(\|A_n\|)$ is a cauchy sequence, because (A_n) is. This means the sequence $(\|A_n\|)$ is bounded, and hence, there is a constant $M \in \mathbb{R}$ such that $M \geqslant \|A_n\|$ for all n. Thus $\forall u \neq 0$

$$\frac{\|A_n(u)\|_W}{\|u\|_V} \leqslant M$$

and by the definition of A, then also

$$\frac{\|A(u)\|_W}{\|u\|_V} \leqslant M$$

which shows that A is bounded.

It remains to show that (A_n) converges to A in operator norm. Since (A_n) is a cauchy sequence, there is given $\varepsilon > 0$ an $N \in \mathbb{N}$ such that

$$\|A_n - A_m\| < \varepsilon \ \forall n, m \geqslant N.$$

This means that

$$\|A_n(u) - A_m(u)\| \leqslant \varepsilon \|u\| \ \forall u \in V.$$

If we let $m \to \infty$, we get that

$$\|A_n(u) - A(u)\| \leqslant \varepsilon \|u\| \ \forall u \in V,$$

which means that $\|A_n - A\| < \varepsilon$. This shows that (A_n) converges to A.

Oppgave 17: 5.4.2

Check that the map A in ex 1, is a linear functional, and that B is a linear operator.

Bevis 38: 5.4.2

Operator A Let $V = C([a,b], \mathbb{R})$ and $A : V \to \mathbb{R}$ defined by

$$A(u) = \int_a^b u(x)dx$$

A linear functional is a linear map such that $A : V \to \mathbb{R}$. Since the definite integral is a function

$$\int_a^b : V \to \mathbb{R}$$

I just need to check that A is linear.

$$\begin{aligned} A(\alpha u + \beta v) = \int_a^b (\alpha u + \beta v)(x)dx = \int_a^b \alpha u(x) + \beta v(x)dx = \int_a^b \alpha u(x)dx + \int_a^b \beta v(x)dx \\ = \alpha \int_a^b u(x)dx + \beta \int_a^b v(x)dx = \alpha A(u) + \beta A(v) \end{aligned}$$

so it is in fact linear.

Operator B

Oppgave 18: 5.4.3

Check that the map D in ex 2 is a linear operator

Bevis 39: 5.4.3

I have the linear operator $D : V \to U$ defined by

$$D(f) = f'$$

where $U = C([0,1], \mathbb{R})$ and $V = \{g : [0,1] \to \mathbb{R} : \text{ g is differentiable with } g' \in U\}$

I have for $f, g \in V$

$$D(f+g) = (f+g)' = (f') + (g)' = D(f) + D(g)$$

And for $\alpha \in \mathbb{K}$

$$D(\alpha f) = (\alpha f)' = \alpha (f)' = \alpha D(f)$$

So D is a linear operator.

Oppgave 19: 5.4.4

Show that the formula

$$\|A\| = \sup\{\|A(u)\|_W : \|u\|_V = 1\}$$

is an equivalent definition of the operator norm

Bevis 40: 5.4.4

I have the operator norm of an $A \in \mathcal{L}(V, W)$ where V and W are normed spaces given by

$$\|A\| = \sup\left\{\frac{\|A(u)\|_W}{\|u\|_V} : u \neq 0\right\}$$

We can rewrite

$$\frac{\|A(u)\|_W}{\|u\|_V} = \frac{1}{\|u\|_V}\|A(u)\|_W = \left\|\frac{1}{\|u\|_V A(u)}\right\|_W = \left\|A\left(\frac{u}{\|u\|_V}\right)\right\|_W = \|A(v)\|_W$$

where $v := \frac{u}{\|u\|_V}$. So

$$\sup\left\{\|A(v)\|_W : v = \frac{u}{\|u\|_V}, u \neq 0\right\} = \sup\{\|A(v)\|_W : \|v\| = 1\}$$

Oppgave 20: 5.4.5

Define $F : C([0,1], \mathbb{R}) \to \mathbb{R}$ by $F(u) = u(0)$. Show that F is a linear functional. Is F continuous?

Bevis 41: 5.4.5

We have for $u, v \in C([0,1], \mathbb{R})$ that

$$F(u+v) = (u+v)(0) = u(0) + v(0) = F(u) + F(v)$$

and for $\alpha \in \mathbb{K}$ that

$$F(\alpha u) = (\alpha u)(0) = \alpha u(0) = \alpha F(u).$$

So F is a linear functional.

Is F continuosu?

F is continuous if its bounded. That is $\exists M \in \mathbb{R}$ such that

$$|F(u)| \leqslant M\|u\| \ \ \forall u \in C([0,1], \mathbb{R})$$

Let $u \in C([0,1], \mathbb{R})$. We have that since $[0,1]$ is a compact set, then the supremum norm of u is finite.

$$\|u\| < \infty$$

So we get that $u(0) \leqslant \|u\|$. Hence

$$|F(u)| = |u(0)| \leqslant \|u\|$$

where $M := 1$. Hence, F is bounded, which implies its continuous.

Oppgave 21: 5.4.7

Check that $\mathcal{L}(V, W)$ is a linear space

Bevis 42: 5.4.7

Let $A, B, C \in \mathcal{L}(V, W)$ and $\alpha, \beta \in \mathbb{K}$. Then we define the algebraic operations

i) $(A + B)(u) = A(u) + B(u)$

ii) $(\alpha A)(u) = \alpha A(u)$

We have eight points to cover

i) $(A + B)(u) = A(u) + B(u) = B(u) + A(u) = (B + A)(u)$

ii) $(A + B)(u) + C(u) = A(u) + B(u) + C(u) = A(u) + (B + C)(u)$

iii) there is a zero operator $\mathbf{0} \in \mathcal{L}(V, W)$ such that $(A + textbf0)(u) = A(u)$. We have that $(A + \mathbf{0})(u) = A(u) + \mathbf{0}(u) = A(u)$

iv) $(A + (-A))(u) = A(u) + (-A)(u) = A(u) - A(u) = \mathbf{0}$

v) $(\alpha(A + B))(u) = (\alpha A + \alpha B)(u) = (\alpha A)(u) + (\alpha B)(u) = \alpha A(u) + \alpha B(u)$

vi) $((\alpha + \beta)A)(u) = (\alpha A + \beta A)(u) = (\alpha A)(u) + (\beta A)(u) = \alpha A(u) + \beta A(u)$

vii) $(\alpha(\beta A))(u) = \alpha(\beta A)(u) = \beta(\alpha A)(u) = (\beta(\alpha A))(u)$

viii) $(1A)(u) = (A)(u) = A(u)$

So $\mathcal{L}(V, W)$ is in fact a linear space.

5.5 Inverse operators and Neumann series

Definisjon 11: 5.5.1

If $A : U \to V, B : V \to W$ are two linear operators, we write BA as their composition

$$BA = B \circ A$$

If $C : U \to U$ is a linear map from a space to the same space, we write

$$C^n = C \circ C \circ ... \circ C$$

for the result of composing C with itself, n times.

Lemma 3: 5.5.2

Assume U, V, W are normed spaces and that $A : U \to V$ and $B : V \to W$ are two bounded linear operators. Then the composition BA is bounded and

$$\|BA\| \leqslant \|B\| \|A\|$$

For $C : U \to U$ we have

$$\|C^n\| \leqslant \|C\|^n$$

Bevis 43: 5.5.2

For any $u \in U$ we have

$$\|BA(u)\|_W \leqslant \|B\| \|A(u)\|_V \leqslant \|B\| \|A\| \|u\|_U$$

and $\|C^n\| \leqslant \|C\|^n$ follows from induction.

If U is a linear space, we let I_U denote the identity operator on U.

$$I_U : U \to U$$

defined by

$$I_U(u) = u \; \forall u \in U$$

Definisjon 12: 5.5.3

Assume V, W are normed spaces. A bounded linear operator AV/toW is invertible if there is a bounded linear operator $B : W/toV$ such that

$$BA = I_V,\ AB = I_W.$$

We call B the inverse operator of A and denote it A^{-1}

Proposisjon 12: 5.5.4

Assume U, V, W are linear spaces, and that $A : U \to V$ and $B : V \to W$ are two ivertible linear operators. Then BA is invertible and $(BA)^{-1} = A^{-1}B^{-1}$

Bevis 44: 5.5.4

$(BA)(BA)^{-1} = BAA^{-1}B^{-1} = BIB^{-1} = BB^{-1} = I_W$

Proposisjon 13: 5.5.5

Assume V and W are vector spaces and that $A : V \to W$ is a bijective, linear operator. THen the inverse $B : W \to V$ is linear

Bevis 45: 5.5.5

$A(B(\alpha u + \beta v)) = \alpha u + \beta v$. But on the other hand,

$$A(\alpha Bu + \beta Bv) = \alpha A(Bu) + \beta A(Bv) = \alpha u + \beta v$$

as A is injective, this means that $B(\alpha u + \beta v) = \alpha Bu + \beta Bv$ and hence, B is linear.

So if a linear map has an inverse, the inverse is automatically linear. But if a bounded linear map has an inverse, then the inverse is not necessarily bounded.

Consider $U = C([0,1], \mathbb{R})$ with supremum norm $\|\cdot\|$, and put $V = \{g : [0,1] \to \mathbb{R} :$ g is differentiable with $g' \in U, g(0) = 0\}$ and give it the supremum norm as well. Define $A : U \to V$ by

$$A(f)(t) = \int_0^t f(s)ds.$$

As $|A(f)(t)| \leqslant \int_0^t |f(s)|\, ds \leqslant \int_0^1 \|f\|\, ds = \|f\|$, A is bounded. If $g \in V$ then

$$g(t) = g(0) + \int_0^t g'(s)ds = \int_0^t g(s)ds = A(g')(t)$$

by the fundamental theorem of calculus, and hence A is surjective ($A(f) = A(d) \iff f = d$). To see this, just note that if $A(f) = A(h)$ then $\int_0^t f(s)ds = \int_0^t h(s)ds\ \forall t$, and if we differentiate both sides, we get $f(s) = h(s)\ \forall s$.

Using the fundamental theorem of calculus again, we see that the inverse of A is the operator $D : V \to U$ given by $D(f) = f'$. The problem is that differentiation is not bounded. To see this, let $f(x) = \sin((nx)$. THen $f'(x) = n\sin(nx)$ and we get $\|f\| = 1$ and $\|Df\| = n$. Since n can be arbitrarily large, D is not bounded.

This problem, doesnt occur if we are in complete spaces, we will see this in theorem 5.7.5. For now, we will compute the inverse of an operator as a power series. The starting point is familiar.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \text{ for } |x| < 1$$

(REMARK:

We have that $\sum_{n=1}^{\infty} x^n = \frac{x}{1-x}$)

for the sum of a geometric series. It turns out that this generalises well to a formula

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n \text{ for } \|A\| < 1.$$

And

$$\frac{A}{I-A} = \sum_{n=1}^{\infty} A^n$$

for $\|A\| < 1$

for operators $A : U \to U$ for U normed space. This is called a *Neumann Series*

Lemma 4: 5.5.6

Assume U, V, W normed spaces, and $A_n : U \to V$ and $B_n : V \to W$ are bounded linear operators for all n. If $(A_n), (B_n)$ converge to A and B respectively in operator norm, then $(B_n A_n)$ converges to BA in operator norm.

Bevis 46: 5.5.6

We have that given $\varepsilon_1, \varepsilon_2$ that $\forall n \geqslant N_1 \in \mathbb{N}$ that

$$\|A_n - A\| < \varepsilon_1$$

and $\forall n \geqslant N_2 \in \mathbb{N}$ that

$$\|B_n - B\| < \varepsilon_2$$

Hence, we have for all $n \geqslant N := max(N_1, N_2)$ that

$$\begin{aligned}\|B_n A_n - BA\| = \|B_n A_n - BA_n + BA_n - BA\| &\leqslant \|B_n A_n - BA_n\| + \|BA_n - BA\| \\ &\leqslant \|A_n\| \|B_n - B\| + \|B\| \|A_n - A\|\end{aligned}$$

Since a vector space is closed (closed under vector addition and multiplication), and $(B_n) \to B$, then $B \in \mathcal{L}(V, W)$ is a bounded linear operator, and since $(A_n) \to A$, then (A_n) is bounded, hence

$$\|A_n\| \|B_n - B\| + \|B\| \|A_n - A\| < M_1 \varepsilon_2 + M_2 \varepsilon_1 := \varepsilon$$

hence, we have proved that given $\varepsilon > 0$ there is a $N \in \mathbb{N}$ such that for all $n \geqslant N$ then

$$\|B_n A_n - BA\| < \varepsilon$$

which proves that $(B_n A_n) \to BA$

Teorem 5: 5.5.7 Neumann Series

Assume U is complete, normed space and $A : U \to U$ is a bounded linear operator with $\|A\| < 1$. Then $I - A$ is invertible and

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$$

where the series converges in operator norm. Moreover, $\left\|(I-A)^{-1}\right\| \leqslant \frac{1}{1-\|A\|}$

Bevis 47: 5.5.7

From thm 5.4.8 the space $\mathcal{L}(U)$ is complete, and the partial sums are

$$S_n = \sum_{j=0}^{n} A^j.$$

Note that

$$\|S_{n+k} - S_n\| = \left\| \sum_{i=0}^{k-1} A^{n+1+i} \right\| \leqslant \left\| A^{n+1} \right\| \sum_{i=0}^{k-1} \|A\|^k \leqslant \frac{\|A\|^{n+1}}{1 - \|A\|}$$

because $S_{n+k} = \sum_{j=0}^{n+k} A^j$ and $S_n = \sum_{j=0}^{n} A^j$ and

$$S_{n+k} - S_n = \sum_{j=0}^{n+k} A^j - \sum_{j=0}^{n} A^j = \sum_{j=n+1}^{n+k} A^j$$

now introduce $i = j - (n+1) \Rightarrow j = n + 1 + i$. When $j = n+1$, then $i = 0$, and when $j = n+k$, then $i = k - 1$. Hence

$$\sum_{j=n+1}^{n+k} A^j = \sum_{i=0}^{k-1} A^{n+1+i}.$$

As $\|A\| < 1$ we can get this expression as small as we want by choosing n large enough. And hence (S_n) is a cauchy sequence. SInce $\mathcal{L}(U)$ is complete, the partial sums must converge to a sum

$$S = \sum_{n=0}^{\infty} A^n.$$

To see that $S = (I - A)^{-1}$, note that by the lemma,

$$\begin{aligned} S\,(I - A) &= \lim_{n\to\infty} S_n\,(I - A) = \lim_{n\to\infty} \left(\sum_{i=0}^{n} A^i \right) (I - A) \\ &= \lim_{n\to\infty} \left(\sum_{i=0}^{n} A^i - \sum_{i=1}^{n+1} A^i \right) = \lim_{n\to\infty} \left(I - A^{n+1} \right) = I \end{aligned}$$

where the last step uses that $\|A\| < 1$. A totally similar calculation shows that $(I - A)\,S = I$. Hence $S = (I - A)^{-1}$.

To estimate the norm, note that

$$\left\| (I - A)^{-1} \right\| = \lim_{n\to\infty} \|S_n\|$$

and that

$$\|S_n\| \leqslant \sum_{j=0}^{n} \|A\|^j \leqslant \frac{1}{1 - \|A\|}.$$

Hence, $\left\| (I - A)^{-1} \right\| = \lim_{n\to\infty} \|S_n\| \leqslant \frac{1}{1-\|A\|}$

Remark:

If we let $k \to \infty$ in the inequality,

$$\|S_{n+k} - S_n\| \leqslant \frac{\|A\|^{n+1}}{1 - \|A\|}$$

and we get $\|S - S_n\| \leqslant \frac{\|A\|^{n+1}}{1-\|A\|}$ and we get excellent control over the rate of convergence. This is important for numerical applications of Neumann series

Korollar 1: 5.5.8

Assume U is complete normed space and $B : U \to U$ is a boudned linear operator, with $\|I - B\| < 1$. Then B is invertible and

$$B^{-1} = \sum_{n=0}^{\infty} (I - B)^n .$$

Moreover, $\left\|B^{-1}\right\| \leqslant \frac{1}{1-\|I-B\|}$

Teorem 6: 5.5.9 Banachs lemma

Assume U is complete normed space and that $C, D : U \to U$ are two bounded linear operators. If C is invertible and $\|C - D\| \leqslant \frac{1}{\|C^{-1}\|}$, then D is invertible and

$$\left\|D^{-1}\right\| \leqslant \frac{\left\|C^{-1}\right\|}{1 - \|C^{-1}\| \|C - D\|}$$

Bevis 48: 5.5.9

Put $B = C^{-1}D$ then $I - B = C^{-1}C - C^{-1}D = C^{-1}(C - D)$ and hence

$$\|I - B\| \leqslant \left\|C^{-1}\right\| \|C - D\| < 1$$

by previous theorem, B is invertible and $\left\|B^{-1}\right\| \leqslant \frac{1}{1-\|I-B\|}$. SInce $B = C^{-1}D$, we have that $D = CB$ and hence by prop 5.5.4, D is invertible with $D^{-1} = B^{-1}C^{-1}$. Also

$$\left\|D^{-1}\right\| \leqslant \left\|B^{-1}\right\| \left\|C^{-1}\right\| \leqslant \frac{1}{1 - \|I - B\|} \left\|C^{-1}\right\| = \frac{\left\|C^{-1}\right\|}{1 - \|I - C^{-1}D\|}.$$

As $\left\|I - C^{-1}D\right\| = \left\|C^{-1}(C - D)\right\| \leqslant \left\|C^{-1}\right\| \|C - D\|$, we get

$$\left\|D^{-1}\right\| \leqslant \frac{\left\|C^{-1}\right\|}{1 - \|C^{-1}\| \|C - D\|}$$

and the theorem is proved.

Oppgave 22: 5.5.1

Assume U, V, W are linear spaces and that $A : U \to V, B : V \to W$ are linear operators. SHow that BA is a linear operator.

Bevis 49: 5.5.1

i) We have for $\alpha \in \mathbb{K}$ and $u \in U$ that

$$(BA)(\alpha u) = B(A(\alpha u)) = B(\alpha A(u)) = \alpha B(A(u)) = \alpha BA(u)$$

ii) For $u, v \in U$ we have that

$$(BA)(u + v) = B(A(u + v)) = B(A(u) + A(v)) = B(A(u)) + B(A(v)) = BA(u) + BA(v)$$

Oppgave 23: 5.5.2

Assume U is a normed space and that $C : U \to U$ is a bounded linear operator. SHow that $\|C^n\| \leqslant \|C\|^n$

Bevis 50: 5.5.2

For $n = 0$ its obvious, because $\left\|1^1\right\| = \|1\|^1$.

$n = k$ we assume our induction hypothesis

$$\left\|C^k\right\| \leqslant \|C\|^k$$

$n = k + 1$ here we have that

$$\left\|C^{k+1}\right\| = \left\|C^k C\right\| \leqslant \left\|C^k\right\| \|C\|$$

and since we assume $\left\|C^k\right\| \leqslant \|C\|^k$ we get that

$$\left\|C^{k+1}\right\| \leqslant \|C\|^k \|C\| = \|C\|^{k+1}$$

and our proof is complete

Oppgave 24: 5.5.3

Assume U, V are normed spaces and that $A : U \to V$ is an invertible linear operator whose inverse A^{-1} is bounded. Show that

$$\|A(u)\|_V \geqslant \frac{1}{\|A^{-1}\|} \|u\|_U \ \ \forall u \in U$$

Bevis 51: 5.5.3

We have for $u \in U$ that

$$\|u\|_U = \left\|A^{-1}(A(u))\right\|_U \leqslant \left\|A^{-1}\right\| \|A(u)\|_V .$$

Hence, we get that

$$\|u\|_U \frac{1}{\|A^{-1}\|} \leqslant \|A(u)\|_V$$

Oppgave 25: 5.5.4

Assume U, V are normed spaces and that $A : U \to V$ is a surjective, bounded linear operator. SHow that if there is a $c > 0$ such that $\|A(u)\|_V \geqslant c \|u\|_U$ for all $u \in U$, then A is invertible.

Bevis 52: 5.5.4

If A is bounded, then its uniformly continuous, and $\|A\| < \infty$. If A is surjective, then $\forall v \in V$ there is a $u \in U$ such that $A(u) = v$. A is invertible when there is a bounded linear operator $B : V \to U$ such that

$$BA = I_U$$

Let $u \in U$. Assume that there is a $c > 0$ such that

$$\|A(u)\|_V \geqslant c \|u\|_U .$$

Have to show that A is invertible. We have to show that A is injective. A is injective if $Au = 0 \iff u = 0$. Since

$$\|Au\|_V \geqslant c \|u\|_U$$

then $\|A(u)\|_V = 0 \geqslant c \|u\|_U$. Since $c > 0$, then $\|u\|_U = 0 \iff u = 0$. So $Ker(A) = \{0\}$.

Since A is injective and surjective, its bijective, hence its invertible. I.e. there is a $A^{-1} : V \to U$ such that $A^{-1}A = I$

Have to now show that its a bounded operator. Introduce $Au = v \Rightarrow u = A^{-1}v$.Substitute this and get

$$\|v\|_V \geqslant c \left\|A^{-1}v\right\|_U .$$

Since $c > 0$, then $\frac{1}{c} > 0$ and

$$\left\|A^{-1}v\right\|_U \leqslant \frac{1}{c} \|v\|_V \ \forall v \in V.$$

Hence, A is invertible.

Oppgave 26: 5.5.5

Assume U is complete normed space. SHow that the set of inverible operators $A : U \to U$ is open in $\mathcal{L}(U)$.

Bevis 53: 5.5.5

A set C is open if for all elements $u \in C$ there is a $r > 0$ such that $B(u, r) \subseteq C$

$$B(u, r) = \{\|u - v\| < r : v \in C\}$$

Since U is complete, then $\mathcal{L}(U)$ is complete as well, i.e. all cauchy sequences converge. That is for (A_n) in $\mathcal{L}(U)$ then given $\varepsilon > 0 \ \exists N \in \mathbb{N}$ such that

$$\|A_n - A_m\| < \varepsilon \ \forall n, m \geqslant N$$

Let $A \in GL(U)$ where $GL(U)$ is the set of invertible operators. This means that A is invertible. Suppose $B \in \mathcal{L}(U)$ satisfies $\|B - A\| < \varepsilon := \frac{1}{2\|A^{-1}\|}$. Then

$$\left\|A^{-1}(B - a)\right\| \leqslant \left\|A^{-1}\right\| \|B - A\| < \left\|A^{-1}\right\| \varepsilon = \frac{1}{2} < 1$$

Define $C := -A^{-1}(B - A)$, since $\|C\| < 1$ then $I - C$ is invertible. But $B = A + (B - A) = A - AA^{-}1(B - A) = A + AC = A(I - C)$. Since A and $I - C$ are invertible, then B is also invertible. Hence $B \in GL(U)$ and $B(A, \varepsilon) \subseteq GL(U)$ which proves that $GL(U)$, the set of invertible operators is open in $\mathcal{L}(U)$.

Oppgave 27: 5.5.6

Assume U is a complete normed space and that $A : U \to U$ is an invertible operator. Show that if (A_n) is a sequence of bounded linear operators converging to A in operator norm, then A_n is invertible for all sufficiently large n, and the resulting sequence $\left(A_n^{-1}\right)$ converges to A^{-1}

Bevis 54: 5.5.6

To first prove that A_n are inversable, we let $\varepsilon > 0$ and $N \in \mathbb{N}$. Then, since $(A_n) \to A$, we have that $\forall n \geqslant N$ that $\|A_n - A\| < \varepsilon$. From Banachs lemma, we have that if A is inversible, and

$$\|A_n - A\| < \frac{1}{\|A^{-1}\|},$$

then A_n is inversable as well. So now we just put $\varepsilon := \frac{1}{\|A^{-1}\|}$. Since we assume A is inversable to begin with, this works just fine. So A_n is inversable for all $n \geqslant N$

Now to prove that the sequence $\left(A_n^{-1}\right) \to A^{-1}$. We have to prove that for the same N, then for all $n \geqslant N$, $\left\|A_n^{-1} - A^{-1}\right\| < \varepsilon_2$ for some other $\varepsilon_2 > 0$.

$$A_n = A + (A_n - A) = A\left(I + A^{-1}(A_n + A)\right)$$

Define $T_n := A^{-1}(A_n + A)$, so $A_n = A(I + T_n)$. Since $(A_n) \to A$ in operator norm, then

$$\|T_n\| = \left\|A^{-1}(A_n - A)\right\| \leqslant \left\|A^{-1}\right\| \|A_n - A\| < \left\|A^{-1}\right\| \varepsilon = \left\|A^{-1}\right\| \frac{1}{\|A^{-1}\|} = 1$$

So $\|T_n\| < 1$ and this makes $I - T_n$ invertible. We get that

$$\left\|(I - T_n)^{-1}\right\| \leqslant \|T_n - I\|^{-1} = \frac{1}{1 - \|T_n\|}.$$

Now compute $A_n^{-1} - A^{-1}$.

$$A_n^{-1} - A^{-1} = A^{-1}(I + T_n)^{-1} - A^{-1} = A^{-1}\left((I + T_n)^{-1} - I\right)$$

We have from neumann seires that

$$\left\|(I + T_n)^{-1} - I\right\| \leqslant \frac{\|T_n\|}{1 - \|T_n\|}.$$

The reason we get this identity, is the following. We have that $\|T_n\| < 1$, and from 5.5.7,

$$(I - T_n)^{-1} = \sum_{k=0}^{\infty} T_n^k = I + T_n^1 + T_n^2 + ...$$

So $(I - T_n)^{-2} - I = \left(I + T_n^1 + T_n^2 + ...\right) - I = T_n^1 + T_n^2 + ... = \frac{\|T_n\|}{1 - \|T_n\|} = \sum_{k=1}^{\infty} \|T_n\|^k$

Since $\|T_n\| \leqslant \left\|A^{-1}\right\| \|A_n - A\| \to 0$, then

$$\begin{aligned}\left\|A_n^{-1} - A^{-1}\right\| = \left\|\left((I + T_n)^{-1} - I\right) A^{-1}\right\| &\leqslant \left\|(I + T_n)^{-1} - I\right\| \left\|A^{-1}\right\| \\ &\leqslant \frac{\|T_n\|}{1 - \|T_n\|} \left\|A^{-1}\right\| \to 0\end{aligned}$$

by the same logic. Hence, there exists a $N \in \mathbb{N}$ such that given $\varepsilon_2 > 0$, then $\forall n \geqslant N$ we have that

$$\left\|A_n^{-1} - A^{-1}\right\| < \varepsilon_2$$

Bevis 55: 5.5.6

Alternatively observe that

$$\left\|A_n^{-1} - A^{-1}\right\| = \left\|A_n^{-1}\left(A - A_n\right)A^{-1}\right\| \leqslant \left\|A_n^{-1}\right\| \|A - A_n\| \left\|A^{-1}\right\| < \left\|A_n^{-1}\right\| \left\|A^{-1}\right\| \varepsilon.$$

Now we just need to prove that $\left\|A_n^{-1}\right\|$ and $\left\|A^{-1}\right\|$ are bounded linear operators. We are guaranteed linearity, but boundedness comes from Banachs lemma, where

$$\left\|A_n^{-1}\right\| \leqslant \frac{\left\|A^{-1}\right\|}{1 - \left\|A^{-1}\right\| \|A - A_n\|}$$

Note that we are able to bound $\left\|A_n^{-1}\right\|$ and $\left\|A^{-1}\right\|$ because we are dealing with complete spaces. This is not generally true!

Oppgave 28: 5.5.7

Assume $a, b \in \mathbb{R}$ with $a < b$ and that $K : [a, b] \times [a, b] \to \mathbb{R}$ is a continuous function. Show that

$$A(y)(t) = \int_a^b K(t, s)y(s)ds$$

defines a bounded linear operator $A : C([a, b], \mathbb{R}) \to C([a, b], \mathbb{R})$ when $C([a, b], \mathbb{R})$ is given the usual supremum norm

Bevis 56: 5.5.7

We have to show that $A : C([a, b], \mathbb{R}) \to C([a, b], \mathbb{R})$ is bounded and linar

Linear

- For $f \in C([a, b], \mathbb{R})$ and $\alpha \in \mathbb{R}$ we have that

$$A(\alpha f)(t) = \int_a^b K(t, s)(\alpha f)(s)ds = \int_a^b \alpha K(t, s)f(s)ds = \alpha \int_a^b K(t, s)f(s)ds = \alpha A(f)(t)$$

- For $f, g \in C([a, b], \mathbb{R})$ that

$$A(f + g)(t) = \int_a^b K(t, s)(f + g)(s)ds = \int_a^b K(t, s)f(s)ds + \int_a^b K(t, s)g(s)ds = A(f)(t) + A(g)(t)$$

Bounded A is bounded if $\|A\| < \infty$ or if $\|A(f)\| \leqslant c\|f\| \ \forall f \in C([a, b], \mathbb{R})$. Since $K : [a, b] \times [a, b] \to \mathbb{R}$ is continuous and defined on a compact set, then

$$\overline{K} = \sup\{|K(t, s)| : t, s \in [a, b]\}$$

and $K(t, x) \leqslant \overline{K}$.

Furthermore, we also have that d

$$|f(t)| \leqslant \sup\{|f(t)| : t \in [a, b]\} =: \|f\|$$

Therefore,

$$\|A(f)(t)\| = \left\|\int_a^b K(t, s)f(s)ds\right\| \leqslant \int_a^b |K(t, s)f(s)|\, ds \leqslant \int_a^b |K(t, s)|\, |f(t)|\, ds$$
$$\leqslant \int_a^b \overline{K}\, \|f\|\, ds = \overline{K}\, \|f\|\, (b - a)$$

Hence, A is bounded, and its linear.

Oppgave 29: Extraoppgave fra Ulrik

Let $A : \ell^\infty \to \ell^\infty$ be the function

$$A\left((a_1, a_2, ...)\right) = (0, a_1, a_2, ...)$$

a) Show that A is a bounded linear operator

Bevis 57: a)

A LINEAR:

For $a, b \in \ell^\infty$ where $a = (a_1, a_2, ...), b = (b_1, b_2, ...)$

$$A(a+b) = A(a_1 + b_1, a_2 + b_2, ...) = (0, a_1 + b_1, a_1 + b_2) = (0, a_1, a_2, ...) + (0, b_1, b_2, ...) = A(a) + A(b)$$

For $a \in \ell^\infty$ and $\alpha \in \mathbb{R}$

$$A(\alpha a) = A(\alpha a_1, \alpha a_2, ...) = (\alpha \cdot 0, \alpha a_1, \alpha a_2, ...) = \alpha(0, a_1, a_2, ...) = \alpha A(a)$$

A BOUNDED: The norm of an infinite vector in ℓ^∞ is given by $\|a\| = \sup\{|a_i| : i \in \mathbb{N}\} =: M$. This is because every vector in ℓ^∞ is bounded by a number.

$$\|A\| = \sup\left\{\frac{\|Aa\|}{\|a\|} : a \in \ell^\infty, a \neq 0\right\}.$$

We have that

$$\|Aa\| = \|A\left((a_1, a_2, ...)\right)\| = \|(0, a_1, a_2, ...)\| \leqslant M$$

and

$$\|a\| \leqslant M$$

So $\|A\| = 1$.

b) Find a bounded linear operator $B : \ell^\infty \to \ell^\infty$ such that $BA = I$

Bevis 58: b)

Define $B : \ell^\infty \to \ell^\infty$ as

$$B(b) = B\left((b_1, b_2, ...)\right) = (b_2, b_3, ...).$$

We then get that

$$BA(a) = B(A(a)) = B(A((a_1, a_2, ...)) = B(0, a_1, a_2, ...) = (a_1, a_2, ...).$$

Hence, $BA(a) = a$, hence $BA = I$.

c) Show that $AB \neq I$

Bevis 59: c)

We have that

$$BA(a) = a$$

but

$$A(B(a)) = A(B((a_1, a_2, ...))) = A((a_2, a_3, ...)) = (0, a_2, a_3, ...).$$

Hence, $AB \neq I$